

C Basics

Course: *Parallel Programming and High Performance Computing*
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Introduction

Goal of this document. This is a collection of materials that you can work through on your own, additionally to the recommended books in the [module description](#). In the exercises throughout this course you will be asked to apply what you have learned in C, so in preparation for the course it will be useful to learn it. We collected accessible online tutorials that will teach you the parts of C that you will need. If you already know a bit of C you might want to check if you have heard about all the concepts. Beyond that you are of course always welcome to learn more about it. Let's get going!

Me: *misses semicolon on one of
my lines*
C compiler:



<https://imgur.com/gallery/t5w1XxX>

Tutorial list: C Programming

The list is ordered to start from scratch so that all concepts that are used in an article have been explained before. Sometimes there are things in the code that have not been introduced, they will appear later if they are important. Make sure to click links within tutorial articles to see the explanation of the subpoints. @ stands for a small quiz to test yourself. After reading the first four sections you will be able to compile and run C programs, and we encourage you to write (and debug) small programs for each section to play around with what you have learned. See also the available [exercises](#). If you are using Windows and did not get around to setting up a Linux Virtual Machine yet we recommend you to use an online compiler like the [GDB online debugger](#) instead of installing gcc on your machine. This online compiler allows for multiple files and compiler flags. Nevertheless, in the long run we want you to use Linux and to be able to compile locally from your command line.

1. Introduction

Why C?

Setting up your computer¹

Hello World @

2. From source code to executable

Preprocessor (until including Examples)

Header Files

3. Primordial Soup

Basic Syntax

Data Types

In case you missed boolean values

Enumeration type

Variables

Constants

Operators

Variable interpretation - Type Casting @

4. Input and Output

Standard Input and Output

Formatted printing

File Management @

5. Control Structures

Conditions – if-else, switch statements @ @

Loops @

6. Organizing Code

Functions @

Scope of Variables

Sharing Variables - Storage Classes

7. Pointers

The NULL Pointer

¹Instead of coding in a text editor you can use an Integrated Development Environment (IDE). They provide functionalities like going to certain places in your code (e.g. "Go to definition of function foo"), code correction proposals, editing compiler settings in a GUI, debugging etc. For the small programs in the exercises it does not make such a difference. Your learning curve in terms of syntax will be higher if you code in a bare editor. On the HPC cluster there won't be any IDE available. But you can transfer files, see the course on Access to the HPC cluster.

8. Data Structures

[Arrays @](#)

[Strings](#)

[Structures @](#)

[User Defined Data Types](#)

9. Memory

[Memory Allocation @](#)

[Heap and Stack](#)

10. [Command Line Arguments](#)²

11. [GCC and GNU Make](#) – Compiling, Linking and Building in more detail

²Nice to know.

Exercises

Programming is learning by doing. On [OPAL](#) we provide you with some toy examples for getting your hands dirty on some code. This also helps experienced programmers who know the concepts but need to learn the specific C syntax.

If the code does not compile don't fret! Check out the output of the compiler. If you forgot a semicolon, a bracket or similar at the end of the expression it is possible that the compiler outputs a line number after the actual error. If you reach the end of your wisdom ask a classmate - two heads are better than one! If you're coding at 3 am and no one replies - [Stack Overflow](#) is your friend.

Attributions

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